



Workshop Series Spring 2026

CAMBAM

March 30th, 2026, 1pm-5pm

Normative modelling using the PCNtoolkit

Organized by Michelle Wang & Johanna Bayer, McGill Univ.

Overview Normative modelling provides a Bayesian framework for characterizing individual variation relative to a reference population, enabling inference beyond group averages. Originally developed in pediatrics (growth charts) and neuroimaging (brain charting), normative models are increasingly relevant across quantitative life sciences, including physiology, genetics, and longitudinal health data, where heterogeneity and uncertainty are central challenges. This workshop introduces the theoretical and computational foundations with hands-on training using [PCNtoolkit](#) and [PyMC](#). We will provide data and work on Jupyter notebooks together, but you can also bring your own data.

Workshop:

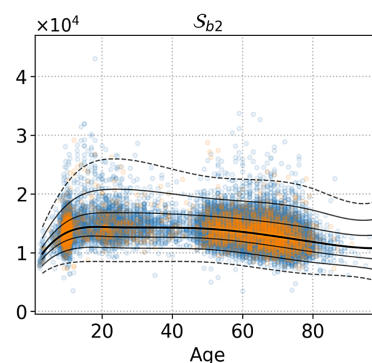
13:00 -13:45 Lecture 1. Theory and concepts

- Why group averages fail for heterogeneous data
- Bayesian foundations: hierarchical models, Markov Chain Monte Carlo, PyMC
- Interpretation of deviation scores and uncertainty

13:45 -15:00 Break

15:00 -17:00 Lecture 2. Practical Session

- Fitting, extending, and transferring models
- Evaluating performance and interpreting deviations
- Extensions: covariates, multi-site, longitudinal modelling



Register now!!

- <https://forms.gle/gkKmsFv5KSLve1vK9>
- This workshop will be held here: Sherbrooke 550, Room 189
- The practical session will require some knowledge of python and your own laptop